intersubjective description of the use-value itself. The German idea of utility was not at all *subjective* because of the socially accepted superior position of *Polizei* over *Oekonomie*. Therefore the conceptual frame of *Oekonomie* could accept the Linnaean way of describing the "use-values of commodities" (on social usefulness in Beckmann's work and in German culture see Troitzsch, 1980). Göttingen University itself, affirmed Heyne in 1788, contributed to the appreciation of German studies on utility (quoted from Marino, 1975, :45).

In conclusion, though this kind of intersubjective description is a very interesting one, it is not in the least a *modern concept* of use-value because use-value in the modern sense is the psychological, sociological or traditional relationship between the social actor and the commodity, that is a *subjective use-value*. Certainly, the Linnaean description of "use-values" does not have this feature and therefore cannot be identified as the basis of modern economics: instead, this kind of description, because of its *scientific* and *intersubjective* character, can be easily identified as the basis of the modern concept of *technology*.

If the proposed analysis on the connection between *Oeconomia* and *Technologie* is accepted, the whole evaluation done up to the present on the works of Linnaeus should be changed. The great botanist is considered, from the point of view of his naturalist achievements, a scholar who terminated a scientific age (see Sachs, 1890, :79–108, especially :80; Lindroth, 1966, :42–43), but from the point of view of technology, Linnaeus opened the modern age. His interest in *utility* is on the path which leads to *Technologie*, that is to a description of productive phenomena which has *no precedent* in the history of world scientific speculation. Unfortunately this point, as well the Linnaean interest in *utility*, has been rather simply defined by scholars to be of an economic kind and therefore connected to a simple *economics* rather than to the beginning of a modern point of view on productive phenomena, i.e. *technology*.

Having clarified the ties between Linnaeus and Beckmann and those between natural history and *Oeconomia*, we will now delineate how it was possible for Beckmann to move from Linnaeus's *Oeconomia* to the threshold of *Technologie*.

In 1767 Beckman rapidly changed his studies to economic matters (applied natural history) and in the same year published a small book with a programme of his future lectures on *Oekonomie* to be given in the winter semester 1767/68 (Beckmann, 1767). The subjects he worked on were not very different from the agricultural kind contained in the *Hausväterliteratur*[14] but the method of exploring the material was experimental: he utilized models of tools, collected seeds and in 1768 built an "economic garden", where it was possible to observe different varieties of cultivated plants and even weeds (Kaufhold, 1989, :55; Beckmann, 1767, :24).

In 1769 Beckmann published a handbook on agricultural matters which earned him the title of *Professor ordinarius Oeconomiae* the following year. This book (*Grundsätze der teuschen Landwirthschaft* = Principles of German Agriculture) had six editions and became very well known. In the opinion of a benevolent scholar, Beckmann "founded the independent science of agriculture" with this work (Klemm, 1870, :554). But more interesting for our purposes are Beckmann's opinions on the relationships between natural sciences and the matters of cameralism which are outlined in a well-ordered classification of cameral sciences beginning with natural history, passing through the science of agriculture (=*Oekonomie*) and *Stadtwirtschaft* (i.e. *Handwerkwissenschaft* [= science of town crafts] and *Handlungwissenschaft* [= science of commerce]) and finally arriving at the science of

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